

Abstract

Chemical cues are a primary source of information for aquatic animals, allowing them to detect predators and respond accordingly. Native crustaceans rely on this heavily for predator detection, but pollutants like petroleum, tire compounds, metals, etc. are becoming increasingly common in estuaries like Puget Sound where the invasive European green crab (*Carcinus maenas*) now preys on these native shore crabs. Because contaminants like copper have been known to impair olfaction in salmon, pollution could plausibly inhibit native antipredator behavior. Whether a realistic pollutant mixture disrupts the ability of the shore crab *Hemigrapsus oregonensis* to chemically detect *C. maenas* has been untested. Here using a 2x2 factorial design crossing pollution and predator cues that pollution, while reducing the metabolic rate of *H. oregonensis*, did not consistently alter its chemosensory antennule-flicking response to the predator. The predator cue effect on metabolism was shown to be unchanged by pollution, and a pollution $\times$ predator interaction on flicking appeared only under high-intensity pollution, where polluted crabs unexpectedly flicked much more, a pattern better explained by the mortality and replacement crabs of that high-intensity group not having been acclimated to the pollutants. With only 3 crabs per treatment, we could not reject the hypothesis that pollution does not affect the stress response of *H. oregonensis* in the presence of *C. maenas*. Sublethal pollution is shown to measurably suppress crab metabolism, but its effect on olfactory predator detection remains unresolved. Pursuit of more data on this subject is important as many marine invertebrates rely on chemoreception for predator detection and by extension, survival. A similar experiment using larger samples with a single contaminant will offer a fairly low-cost framework for testing how pollutants shape antipredator behavior.

Introduction

The European green crab (*Carcinus maenas*), known as one of the worst invasive species in the entire world, has spread throughout the waters in and surrounding Washington since 2016 at the very latest (Washington Department of Fish and Wildlife, n.d.; Washington Sea Grant Crab Team, 2025). Naturally there is overlap in territory with native creatures like shore crabs (genus *Hemigrapsus*) and in these zones the green crabs are effective predators and consume the smaller shore crabs as well as compete for resources (Colnar & Landis, 2007; Ens et al., 2021; Jensen et al., 2002). For native shore crabs like *Hemigrapsus oregonensis*, avoiding green crabs is a matter of life and death.

H. oregonensis detects predators through chemoreception via flicking the olfactory antennules to guide their anti-predator behavior (Symons, 1964; Waldrop, 2013; Waldrop et al., 2015). Since predator detection raises activity and metabolic rate (Birch et al., 2025), antennule-flicking data and metabolic rate data can be used as proxies for stress response from predator detection, to tell biologists if a predator is detected by the crab.

It is known that urban runoff contains several chemicals that impair chemoreception: copper, tire particles, herbicides, petroleum products, etc. Copper, which is common in stormwater, is known to disrupt peripheral olfactory pathways and degrade predator detection in

salmon (Baldwin et al., 2003; Sandahl et al., 2007). Similar studies have shown the same is true for crabs, specifically with *H. oregonensis* exhibiting increased predator-risk behavior when exposed to runoff-based stressors (Birch et al., 2025; Peters et al., 2017) as well as impaired metabolism (Cappello et al., 2017) suggesting that pollutants would affect the measurable metabolism and chemoreception effort of shore crabs.

Whether a fairly realistic Puget Sound pollutant mixture disrupts *H. oregonensis*' ability to detect *C. maenas* has not yet been tested, until we quantified the metabolic and chemosensory response of *H. oregonensis* to *C. maenas* chemical cues across a 2x2 factorial of pollution and predator-cue exposure. We tested the H_0 that pollution does not affect the stress response of *H. oregonensis* in the presence of *C. maenas*; with an H_a , that polluted water alters that response; and a sub-alternative hypothesis that *H. oregonensis* detects *C. maenas* less effectively in polluted water.

Methods

Native shore crabs (*Hemigrapsus oregonensis* and *H. nudus*) were collected from local intertidal zones and held in aerated 35 ppt artificial seawater; each crab's mass, size, pattern and sex were recorded for continuity and scaling. 2x2 factorial design of pollution (absent/present) and predator cue exposure (absent/present) is used to see how the change in stress response from predator cue exposure differs in clean vs polluted environments. Crabs were split between a clean tub (Jars 1A, 2A) and a polluted tub (Jars 1B, 2B) at three crabs per perforated jar, giving four treatments: control (1A), pollution (1B), predator cue (2A), and pollution + predator cue (2B), allowing for chemical exchange without giving visual cues. To prevent cross-contamination, all polluted crabs, *C. maenas* and *grapsids*, were maintained as a separate system. Crabs acclimated a week before data collection.

To simulate intensely polluted environments that could exist in Puget Sound, a mixture was prepared in 2L of 35 ppt seawater: motor oil (100 $\mu\text{g/L}$), organic herbicide (100 $\mu\text{g/L}$), tire-wear pellets (6.849 g), and a 3x3 aluminum sheet coated with two layers of copper antifouling paint). It was left to leech into the tanks and mix well before the 1B and 2B crabs were introduced. Clean jars (1A and 2A) were placed in an identical tank full of 2L of 35 ppt seawater. We allowed these crabs to acclimate for a week and this mixture proved too toxic and killed 2/3 of the pollutant group crabs, leading us to cut the concentrations by half. This yielded two different pollution intensities that are analyzed separately: high (full-strength) and low (half-strength).

On data collection days, the non-predator jars (1A and 1B) are removed from each environment, and the green crab is then placed in the environment (not the jars). All jars were opaque ensuring responses reflected chemoreception rather than visual detection (Waldrop et al., 2015). These crabs are then left to stew in the environment for about 25 minutes. During this time, we collected data on the 1A and 1B treatments. To quantify chemosensory activity, we measured antennule-flicking rate. To do so we filmed the crabs for one minute, and using a clicker, hand counted each crab followed by standardizing for mass and then averaging for a mean antennule-flicking rate for a jar. Metabolic rate was estimated by a resazurin assay, in

which respiring tissue reduces resazurin to create a comparatively fluorescent resorufin with the fluorescence increasing proportionally to cumulative respiration (N. Ozguner 2026). One crab per jar was placed in a vessel of resazurin and fluorescence was read at T0 (0 min), T1 (30 min), and T2 (60 min). Each crab's readings were standardized to its own T0 and, because mass varied ($\approx 1.0\text{--}4.1$ g), to body mass; standardized T2 fluorescence served as the metabolic response. For each response (standardized T2 fluorescence; flicks per minute) and each pollution intensity, a two-way ANOVA (Type II sums of squares, $\alpha = 0.05$) tested the main effects of pollution and predator cue and their interaction. Means are reported $\pm$ SD. Sample sizes were $n = 3$ per treatment. All analyses were performed in R.

Results

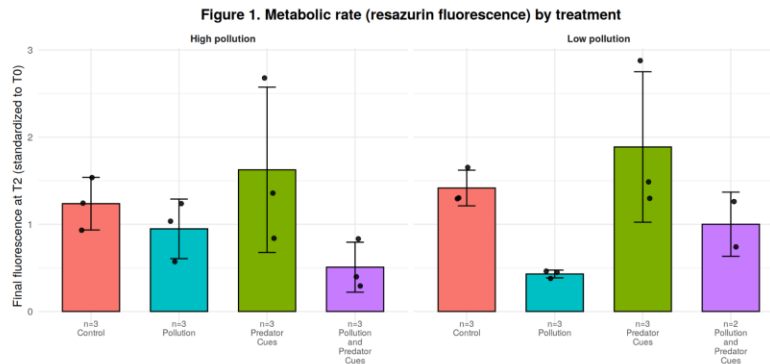


Figure 1. Metabolic rate (resazurin fluorescence at T2, standardized to T0) of *Hemigrapsus oregonensis* by treatment, under high (left) and low (right) pollution. Treatments: Control, Pollution, Predator Cues (*Carcinus maenas*), and both. Bars = means, error bars = ± 1 SD, points = individual crabs, n below each bar.

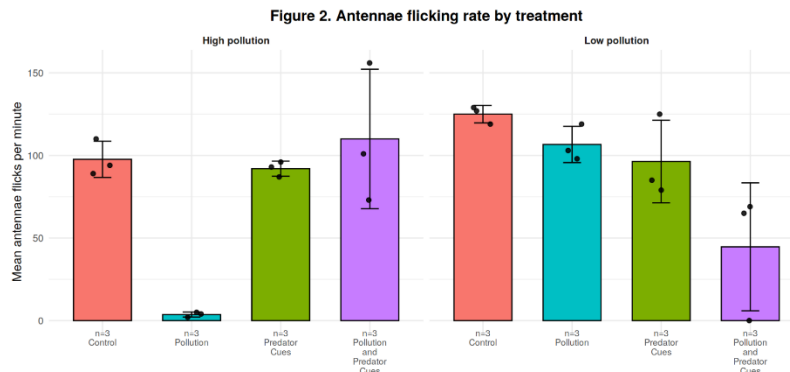


Figure 2. Antennae-flicking rate (flicks min^{-1}) of *H. oregonensis* by treatment, under high (left) and low (right) pollution. Treatments: Control, Pollution, Predator Cues (*C. maenas*), and both. Bars = means, error bars = ± 1 SD, points = individual crabs, n below each bar.

Response (pollution level)	Predator effect, no pollution (2A – 1A)	Predator effect, with pollution (2B – 1B)	Difference between effects	Interaction <i>F</i> (df)	<i>p</i>
Metabolic rate (high)	+0.39	–0.44	–0.83	1.73 (1, 8)	.225
Metabolic rate (low)	+0.47	+0.57	+0.10	0.03 (1, 7)	.874
Antennae flicking (high)	–5.7	+106.3	+112.0	19.53 (1, 8)	.002

Antennae flicking (low)	-28.7	-62.0	-33.3	1.47 (1, 8)	.261
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Table 1. Difference-of-differences test for whether pollution alters the predator-cue response of *H. oregonensis*. The predator effect (predator-exposed – unexposed) is compared without pollution (2A – 1A) and with it (2B – 1B); the pollution × predator interaction tests whether these two effects differ. Estimates come from two-way factorial ANOVAs (Type II) of metabolic rate (standardized fluorescence units) and antennae-flicking rate (flicks min⁻¹). Bold P-values are significant at $P < 0.05$.

Pollution and predator cue interacted to shape antennae-flicking, the response most directly tied to chemosensory predator detection (Table 1). Under high pollution, adding the predator barely changed flicking in clean water (-5.7 flicks/min) but drastically increased it in polluted water (+106.3), a significant difference between the two predator effects (interaction $F(1, 8) = 19.53$, $p = .002$); under low pollution the predator depressed flicking more in polluted than in clean water (-62.0 vs. -28.7), but this difference was not significant ($p = 0.261$). For metabolic rate, the predator effect did not differ significantly between clean and polluted water at either intensity ($p = .225$ and $.874$). Within the broader pattern of main effects, pollution exposure reduced metabolic rate of both predator and non-predator groups (significant at low pollution levels, $F(1, 7) = 9.79$, $p = .017$; marginal at high levels, $F(1, 8) = 4.99$, $p = .056$), and both pollution and predator cue independently shifted flicking at both intensities (all main-effect $p \leq .035$).

Discussion

In the high pollution treatment, it appears that both polluted crab groups (non-predator and predator) had lower average metabolic rates than their unpolluted counterparts indicating the polluted crabs were respiring less. This change in the adjusted low-pollution treatment was very statistically significant as found in the statistical analysis, however the significance of the polluted group in the high-pollution environment significance is lessened by a p-value that only approaches statistically significant ($p = 0.056$). This can likely be attributed to the fact that four crabs, 1 from 2B and all three in 1B, perished over the week of pollution exposure. Other than the general takeaway that metabolic rate of *Hemigrapsus* is depressed under polluted conditions, no other metabolic rate data can be considered significant, meaning we cannot reject the null from this data, as the predator effect on metabolic rate is insignificant regardless of environment, polluted or clean ($p = 0.225$, 0.874).

In the adjusted low-pollution environment, the predator exposed group (2B) had the lowest flicking rate of all at 44.7 flicks/min which by itself fits the idea of pollution impairing predator detection, but when examining the difference in change of antennule-flicking rate after exposure to *Carcinus maenas* between polluted and unpolluted groups, it was shown that while there was a decrease in antenna flicking, it was shown to be statistically insignificant ($p = 0.261$). However, in the high-pollution environment when comparing the difference in change of antennule-flicking rate after predator exposure between the clean and polluted groups, there was a large, very significant difference in the change of antennule-flicking rate. Polluted crabs have a much greater change in their rate compared to their non-polluted counterparts ($p = 0.02$), which by comparison have almost no antennule flicking (difference of 112 flicks/min on average). This large increase would indicate that the polluted crabs would be “smelling” more post predator

exposure in polluted waters rather than clean waters which is counter to the sub hypothesis of *H. oregonensis* ability to detect *C. maenas* deteriorating. But this likely isn't the case, as this data has issues that cannot be reflected in the statistical analysis. The high-pollution environment data is highly unreliable since 4/6 of the crabs used for the polluted data died, and were replaced meaning many of the crabs during that data collection were not acclimated to the polluted environment and are thus not comparable to the low-pollution environment. Unfortunately, none of the data from the low-pollution environment regarding antennule flicking is statistically relevant.

The general low statistical relevance overall throughout the results could likely be attributed to the lack of statistical power; our sample size for each treatment group was only 3 crabs which makes the results with less significant p-values much harder to interpret. Absence of significance doesn't necessarily mean absence of effect, especially with such a low sample size. There is also the issue of using a pollutant cocktail rather than one type of pollutant; any measurable effect cannot be attributed to any one contaminate. Overall sample size and experimental design limit the reliability of any results produced. It is due to all of these compounding factors that the null-hypothesis of pollution not affecting the stress response of *H. oregonensis* in the presence of *C. maenas*, cannot be rejected, and the result of the experiment is largely inconclusive.

As the European green crabs invade further throughout the Salish sea, pollution could be making the native hairy shore crabs' easier prey and we would still continue to be none the wiser, which is why a more carefully planned and measured follow up experiment is necessary. There are studies that have been done that are very similar to this, even one with the same species with a runoff chemical as a variable that impacts the anti-predator behavior of *H. oregonensis* (Peters et al., 2017). Choosing a single chemical seems to be a much better way to approach an experiment of this style, another study very similar to this project in its goal, found that copper inhibited salmon's olfactory ability to sense predators (Sandahl et al., 2007 & Baldwin et al., 2003), so copper would be a good runoff chemical to start with given its measured harm and how common of a pollutant it is.

Overall pollution was shown to have slightly lowered the metabolism of these crabs and it is inconclusive how pollution affects *Hemigrapsus*' stress response to the chemical presence of *C. maenas*. These invasive green crabs are still running rampant and it is still largely unknown how pollution may be assisting these nasty crabs from destroying our native shore crab populations. This is why a follow up experiment with real life pollutant levels with one type of pollutant with more time to prep the setup would allow for the trend of the interaction between predator detection and pollution.

References

Baldwin, D. H., Sandahl, J. F., Labenia, J. S., & Scholz, N. L. (2003). Sublethal effects of copper on coho salmon: Impacts on nonoverlapping receptor pathways in the peripheral olfactory nervous system. *Environmental Toxicology and Chemistry*, 22(10), 2266–2274.

- Birch, A., Cox, K. D., Murchy, K. A., Emry, S., & Harley, C. D. G. (2025). Shipping noise tolerance in invertebrates: A case study of the shore crab *Hemigrapsus oregonensis*. *PLOS ONE*, 20(8), e0329098. <https://doi.org/10.1371/journal.pone.0329098>
- Cappello, T., Maisano, M., & Mauceri, A. (2017). Neurobehavioral and metabolic effects of manganese (Mn) exposure in the Mediterranean crab *Pachygrapsus marmoratus*. *Aquatic Toxicology*, 187, 127–135.
- Colnar, A. M., & Landis, W. G. (2007). Conceptual model development for invasive species and a regional risk assessment case study: The European green crab, *Carcinus maenas*, at Cherry Point, Washington, USA. *Human and Ecological Risk Assessment*, 13(1), 120–155.
- Ens, N. J., Lim, E. G., Howard, B. R., & Eastham, T. M. (2021). A comparison of the predatory impacts of an invasive and native crab species using a functional response approach. *Biological Invasions*, 23(7), 2329–2336.
- Jensen, G. C., McDonald, P. S., & Armstrong, D. A. (2002). East meets west: Competitive interactions between green crab *Carcinus maenas*, and native and introduced shore crab *Hemigrapsus* spp. *Marine Ecology Progress Series*, 225, 251–262. <https://doi.org/10.3354/meps225251>
- Peters, J. R., Granek, E. F., de Rivera, C. E., & Rollins, M. (2017). Prozac in the water: Chronic fluoxetine exposure and predation risk interact to shape behaviors in an estuarine crab. *Ecology and Evolution*, 7(21), 9151–9161. <https://doi.org/10.1002/ece3.3453>
- Sandahl, J. F., Baldwin, D. H., Jenkins, J. J., & Scholz, N. L. (2007). A sensory system at the interface between urban stormwater runoff and salmon survival. *Environmental Science & Technology*, 41(8), 2998–3004. <https://doi.org/10.1021/es062287r>
- Symons, P. E. K. (1964). Behavioral responses of the crab *Hemigrapsus oregonensis* to temperature, diurnal light variation and food stimuli. *Ecology*, 45(3), 580–591. <https://doi.org/10.2307/1936110>
- Waldrop, L. D. (2013). Ontogenetic scaling of the olfactory antennae and flicking behavior of the shore crab, *Hemigrapsus oregonensis*. *Chemical Senses*, 38(6), 541–550. <https://doi.org/10.1093/chemse/bjt024>
- Waldrop, L. D., Hann, M., Henry, A. K., Kim, A., Punjabi, A., & Koehl, M. A. R. (2015). Ontogenetic changes in the olfactory antennules of the shore crab, *Hemigrapsus oregonensis*, maintain sniffing function during growth. *Journal of the Royal Society Interface*, 12(102), 20141077. <https://doi.org/10.1098/rsif.2014.1077>
- Washington Department of Fish and Wildlife. (n.d.). *European green crab (Carcinus maenas)*. Retrieved May 2026, from <https://wdfw.wa.gov/species-habitats/invasive/carcinus-maenas>
- Washington Sea Grant Crab Team. (2025). *2024 green crab status summary: Part 1 (Inland)*. <https://wsg.washington.edu/2024-green-crab-status-summary-part-1-inland/>